Correlation Matrix and Analysis: Student Depression Dataset

1. Introduction

This report presents a comprehensive correlation analysis of the Student Depression Dataset, identifying the key relationships between variables and their association with depression. This analysis is part of Task 11 (T11) in our project plan scheduled for April 12-14, 2025.

2. Correlation with Depression (Target Variable)

The table below shows the correlation coefficients between each variable and the target variable (Depression):

Variable	Correlation with Depression	Strength	Direction
Suicidal Thoughts	0.5463	Strong	Positive
Overall Stress (engineered)	0.5508	Strong	Positive
Academic Pressure	0.4748	Moderate	Positive
Financial Stress	0.3636	Moderate	Positive
Work/Study Hours	0.2086	Weak	Positive
Academic × Satisfaction (engineered)	0.2003	Weak	Positive
Age	-0.2264	Weak	Negative
Dietary Habits	-0.2066	Weak	Negative
Study Satisfaction	-0.1680	Weak	Negative
Sleep Duration	-0.0867	Very Weak	Negative
Family History of Mental Illness	0.0534	Very Weak	Positive
CGPA	0.0222	Very Weak	Positive
Job Satisfaction	-0.0035	Negligible	Negative
Work Pressure	-0.0034	Negligible	Negative
JobSat × WorkPressure (engineered)	-0.0020	Negligible	Negative

2.1 Key Findings

1. **Strongest Positive Correlations with Depression:**
    - Suicidal Thoughts (0.5463): Students reporting suicidal thoughts are much more likely to be classified as depressed
    - Overall Stress (0.5508): The engineered feature combining multiple stress factors shows the strongest correlation

- Academic Pressure (0.4748): Higher academic pressure strongly correlates with higher depression rates

2. **Strongest Negative Correlations with Depression:**
   - Age (-0.2264): Younger students tend to have higher depression rates
   - Dietary Habits (-0.2066): Healthier dietary habits are associated with lower depression rates
   - Study Satisfaction (-0.1680): Higher satisfaction with studies correlates with lower depression rates

3. **Variables with Minimal Correlation:**
   - Work Pressure (-0.0034): Surprisingly, work pressure shows negligible correlation with depression
   - Job Satisfaction (-0.0035): Similarly, job satisfaction shows minimal relationship with depression
   - CGPA (0.0222): Academic performance as measured by CGPA has very little correlation with depression

3. Correlation Matrix for Numerical Variables

3.1 Main Correlation Matrix

The correlation matrix below shows the relationships between all numerical variables:

Variable	Age	Academic Pressure	Work Pressure	CGPA	Study Satisfaction	Job Satisfaction	Work/Study Hours	Financial Stress
Age	1.000	-0.076	0.002	0.005	0.009	0.000	-0.033	-0.095
Academic Pressure	-0.076	1.000	-0.022	-0.022	-0.111	-0.025	0.096	0.152
Work Pressure	0.002	-0.022	1.000	-0.051	-0.021	0.771	-0.005	0.002
CGPA	0.005	-0.022	-0.051	1.000	-0.044	-0.054	0.003	0.006
Study Satisfaction	0.009	-0.111	-0.021	-0.044	1.000	-0.022	-0.036	-0.065
Job Satisfaction	0.000	-0.025	0.771	-0.054	-0.022	1.000	-0.005	0.005
Work/Study Hours	-0.033	0.096	-0.005	0.003	-0.036	-0.005	1.000	0.075
Financial Stress	-0.095	0.152	0.002	0.006	-0.065	0.005	0.075	1.000

3.2 Key Correlations Between Variables

1. **Strong correlations (|r| > 0.5):**
   - Work Pressure and Job Satisfaction (0.771): This strong correlation suggests these variables are measuring related constructs
2. **Moderate correlations (0.3 < |r| < 0.5):**
   - None of the variable pairs show moderate correlations with each other
3. **Weak correlations (0.1 < |r| < 0.3):**
   - Academic Pressure and Financial Stress (0.152): Students with higher academic pressure tend to report slightly higher financial stress
   - Academic Pressure and Study Satisfaction (-0.111): Higher academic pressure is weakly associated with lower study satisfaction
4. **Notable absence of correlations:**
   - CGPA shows minimal correlation with all other variables, including Academic Pressure (-0.022) and Study Satisfaction (-0.044)
   - Age has weak correlations with most variables, suggesting that age-specific effects on depression are not mediated through these factors

4. Correlation Between Academic Factors

Variable	Academic Pressure	Study Satisfaction	CGPA	Work/Study Hours
Academic Pressure	1.000	-0.111	-0.022	0.096
Study Satisfaction	-0.111	1.000	-0.044	-0.036
CGPA	-0.022	-0.044	1.000	0.003
Work/Study Hours	0.096	-0.036	0.003	1.000

4.1 Key Insights on Academic Factors

1. **Academic Pressure and Study Satisfaction** show a weak negative correlation (-0.111), suggesting that as academic pressure increases, study satisfaction tends to decrease slightly.
2. **CGPA is minimally correlated with academic pressure** (-0.022), indicating that academic performance and perceived pressure are largely independent factors in this population.
3. **Work/Study Hours and Academic Pressure** show a very weak positive correlation (0.096), suggesting that students who study more hours perceive slightly higher academic pressure.
4. **Overall, academic factors show weak inter-correlations**, suggesting they capture different aspects of the academic experience rather than being redundant measures.

5. Correlation Between Stress Factors

Variable	Academic Pressure	Financial Stress	Sleep Duration	Dietary Habits	Work/Study Hours
Academic Pressure	1.000	0.152	-0.043	-0.089	0.096
Financial Stress	0.152	1.000	-0.004	-0.087	0.075
Sleep Duration	-0.043	-0.004	1.000	-0.001	-0.028
Dietary Habits	-0.089	-0.087	-0.001	1.000	-0.029
Work/Study Hours	0.096	0.075	-0.028	-0.029	1.000

5.1 Key Insights on Stress Factors

1. **Academic Pressure and Financial Stress** show the strongest correlation among stress factors (0.152), suggesting a weak relationship between these two types of stress.
2. **Sleep Duration and Dietary Habits** show almost zero correlation (-0.001), indicating these lifestyle factors vary independently of each other.
3. **Dietary Habits correlate negatively with both Academic Pressure (-0.089) and Financial Stress (-0.087)**, suggesting that students under higher stress may have slightly poorer dietary habits.
4. **Sleep Duration shows minimal correlation with most stress factors**, suggesting that sleep patterns may be influenced by factors outside the scope of this study.

6. Correlation Matrix for Engineered Features

Variable	Overall_Stress	Academic_x_Satisfaction	JobSat_x_WorkPressure
Overall_Stress	1.000	0.427	0.009
Academic_x_Satisfaction	0.427	1.000	-0.014
JobSat_x_WorkPressure	0.009	-0.014	1.000

6.1 Key Insights on Engineered Features

1. **Overall_Stress and Academic_x_Satisfaction** show a moderate positive correlation (0.427), indicating that these engineered features capture related but distinct aspects of the student experience.
2. **JobSat_x_WorkPressure** shows minimal correlation with other engineered features, suggesting it captures a distinct dimension that is relatively independent.

3. **Engineered features Overall_Stress (0.5508) and Academic_x_Satisfaction (0.2003)** both have stronger correlations with depression than some of their component variables, indicating successful feature engineering.

7. Implications for Predictive Modeling

Based on this correlation analysis, the following recommendations can be made for predictive modeling:

1. **Key Predictor Variables**: The variables with the strongest correlations with depression should be prioritized in model building:
   - Suicidal Thoughts (0.5463)
   - Overall Stress (0.5508)
   - Academic Pressure (0.4748)
   - Financial Stress (0.3636)
   - Age (-0.2264)
2. **Feature Engineering**: The engineered feature "Overall_Stress" shows the strongest correlation with depression (0.5508), stronger than any individual stress measure. This suggests that combining related variables can create more powerful predictors.
3. **Multicollinearity Consideration**: Work Pressure and Job Satisfaction show a high correlation (0.771), suggesting potential multicollinearity. Consider using only one of these variables in predictive models or employ techniques that handle multicollinearity.
4. **Variable Selection**: Variables with negligible correlation with depression (CGPA, Job Satisfaction, Work Pressure) could potentially be excluded from initial models to reduce dimensionality and avoid overfitting.
5. **Interaction Effects**: Given the weak to moderate correlations between some variables (e.g., Academic Pressure and Financial Stress), consider modeling interaction effects to capture their combined impact on depression.
6. **Categorical Variables**: The strong correlation between Suicidal Thoughts and Depression (0.5463) indicates this binary variable should be included in predictive models, despite being categorical in nature.

8. Summary of Key Findings

1. **Suicidal thoughts and Overall Stress** are the strongest correlates of depression among students, with correlation coefficients above 0.54.
2. **Academic Pressure and Financial Stress** show moderate positive correlations with depression (0.47 and 0.36 respectively), suggesting these are significant stressors that may contribute to depression.

3. **Age shows a negative correlation with depression** (-0.23), confirming that younger students are more likely to experience depression, a finding that aligns with the demographic analysis.
4. **Lifestyle factors** like dietary habits (-0.21) and sleep duration (-0.09) show weak negative correlations with depression, indicating healthier lifestyle choices are associated with lower depression rates.
5. **Academic performance (CGPA)** shows almost no correlation with depression (0.02), suggesting that depression affects students across all performance levels.
6. **Most variables show weak inter-correlations**, indicating they capture different dimensions of student experiences and stressors rather than being redundant measures.
7. **The engineered feature Overall_Stress** demonstrates the power of feature engineering, as it shows a stronger correlation with depression than any of its component variables.

These correlation findings provide valuable insights for subsequent modeling tasks and can help guide intervention strategies for addressing student depression by targeting the most significant correlating factors.